Nature-Based Solutions for Urban Heat Literature Review: Can existing urban vegetation measurably reduce heat exposure, and under what environmental conditions is the cooling effect strongest?

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Introduction

Urban areas (cities) are usually hotter than the areas around them because of dense development, sealed surfaces, and waste heat from human activity (Peng et al.; Manoli et al.). Because of this, nature-based solutions such as trees, parks, and other vegetated spaces are often suggested by environmental scientists as relatively low-cost ways to reduce urban heat. Since vegetation can lower temperatures through shade and evapotranspiration, areas with more vegetation would normally be expected to experience less heat (Bowler et al.; Iungman et al.).

However, the cooling effect of vegetation is not always very strong. Research studies show that its effectiveness can vary depending on factors such as climate, vegetation type, spatial arrangement, and the way that heat is measured (Guo et al.; Borah et al.). This literature review therefore focuses on answering two main questions: 1) does existing urban vegetation actually reduce heat exposure, and 2) under what conditions is this cooling effect strongest? To answer these questions, we consider large-scale satellite studies, regional studies from Europe and the United States, and local research from India and other tropical cities. Rather than looking only at how much green space is there in a city, we will also consider how well that vegetation performs in cooling the ecosystem.

Review of Existing Studies

What is measured

Earlier studies on urban vegetation cooling relied on in-situ temperature measurements. Bowler et al. conducted a meta-analysis of studies on parks and street trees and reported an average daytime cooling effect of about 0.9°C in air temperature. However, differences in study design made direct comparisons between various locations difficult. Later on, satellite thermal data made it possible to compare a much larger number of cities. For eg, this allowed Peng and his

colleagues to study surface urban heat island intensity across 419 cities using MODIS land-surface temperature satellite sensor.

Near-surface air temperature was another tool used for measurement. Venter et al. compared data from more than 50,000 crowdsourced weather stations with satellite measurements across 342 European urban clusters. They found that satellite measurements produced heat-island estimates roughly six times larger than estimates based on air temperature. Du et al. found a similar difference when measuring vegetation cooling efficiency. Borah et al. used the humidity-adjusted heat index instead, which takes perceived heat into account rather than temperature alone. This means that the apparent cooling effect of vegetation can change depending on which measure of heat is used, and in some cases the direction of the effect may also change.

What cools, and by how much

Across research in this domain, most studies found that areas with more tree cover were cooler than areas dominated by grass and other low vegetation. Schwaab et al. analysed high-resolution land-surface temperature data from 293 European cities and found that green spaces without trees produced around two to four times less cooling than areas with tree cover. Kim et al. found a similar pattern in a global synthesis of 265 air-temperature measurements from 58 studies. Their results showed that the cooling effect of trees was around two to three times greater than the cooling associated with grass, green roofs, and green walls. Yang et al. examined this relationship across 510 cities using the concept of cooling efficiency, the reduction in land-surface temperature for a given increase in tree cover.

Tree cover alone does not explain the cooling effect. The way trees are spread out spatially across a city also matters. Zhou et al. compared Baltimore with Sacramento, which has a hotter and drier climate. In Baltimore, the percentage of tree cover was a stronger predictor of land-surface temperature, whereas in Sacramento the spatial configuration of vegetation played a larger role. The strength and even the direction of some relationships differed between the two cities. At a smaller scale, Shah et al. studied green spaces in Bengaluru and found that they reduced surface temperature by about 2.2°C, with cooling effects extending to approximately 350 metres from the green space.

Where and when cooling weakens or reverses

This is where the literature has changed most. Zhao et al. established the strong influence of background climate on heat-island intensity and noted the mitigation potential of albedo management, while Manoli et al. said that greening is more efficient in drier regions. Yu et al. supported that irrigation strengthened grassland cooling in hot-dry conditions. Schwaab et al., however, said the opposite gradient within Europe, with weaker tree cooling in the drier south. Two recent studies show us that these are not just simple opposing results. Guo et al. assessed 761 megacities and found that cooling weakens as aridity increases and that in 22% of cities receiving under 1000 mm of annual precipitation, vegetation produced net warming, which was driven particularly by grassland and cropland, where lower albedo and reduced heat storage surpass limited evapotranspiration; under extreme heat, trees failed to cool 25% of cities. Borah et al. analysed 138 Indian cities and saw reversal in the opposite way, where high canopy activity in humid, poorly ventilated cores raises the Heat Index. Ibsen et al. add another axis, finding that if aridity increased nighttime air cooling would too, from 1.4 °C in Miami to 5.6 °C in Las Vegas. Meili et al. explained how different factors, such as sunlight, water loss from plants, and wind, affect cooling. They also showed that the importance of each factor changes with the climate and the time of day.

Who receives the cooling

City-mean cooling does not say much about exposure. Chakraborty et al. found poorer neighbourhoods hotter in 72% of 25 cities examined, with vegetation density the strongest intra-urban driver. Chen et al. quantified a threefold greenspace exposure gap between Global South and Global North cities, with exposure inequality almost twice as high in the South. Li et al. extended this from access to performance, finding that Global South cities deliver roughly 70% of the cooling capacity of Global North cities. Iungman et al. estimated the mortality avoidable by raising tree coverage to 30% across 93 European cities.

Critical Discussion and Research Gap

The studies reviewed here agree less than what the literature's framing suggests. Trees cool more than grass and vegetation lowers daytime surface temperature in most temperate and humid-temperate cities. Beyond these broad patterns, the results become much less consistent. This variation reflects real differences in environmental conditions than just differences in methodology. At both ends of the moisture gradient, vegetation can provide less cooling than

expected or even contribute to higher heat exposure. For example, in arid cities, higher albedo and heat storage effects can offset the limited cooling provided by evapotranspiration. In humid and densely built cities, additional moisture can increase the perceived heat even when air temperature declines. The fact whether vegetation appears to cool or warm an urban environment therefore depends on the background climate, time of the day, urban form, and the temperature measure being used.

Two important gaps emerge from this literature. First is that most studies rely on daytime satellite derived land surface temperature(LST). While it is useful for capturing spatial patterns, LST does not directly represent the heat experienced by people and cannot adequately capture situations in which increased humidity offsets lower air temperature. Second is that most existing evidence is concentrated largely in Europe, China and North America. This is particularly important because of recent evidence of climate dependent reversals from global analyses of aridity and from multi-city research in India. A useful next step would be to examine whether the proposed humidity mechanisms also occur across tropical regions of Asia and Africa and if its effects vary with built density and street level morphology. Looking at humidity, urban form and population alongside temperature could give us a better idea of how vegetation actually affects the heat people experience than just using city wide surface temperature alone.

Conclusion

Through the literature we can conclude that tree canopy generally outperforms grass and hence the existing urban vegetation gives a cooling effect, but this cooling is no longer universal. We can observe this effect by measuring temperatures but it is worth keeping in mind that land-surface estimates are larger (generally) than air-temperate estimates. For this, the most dominant method is Satellite LST, which is analysed across many cities. Nowadays, researchers are combining this with other data such as crowd sourced air temperature, reanalysis climate data and gridded population data. Wherever tree cover is richer and there is enough moisture to support transpiration, cooling is the strongest. But the effect of arrangement of trees differs from city to city and depends on the local climate and characteristics of the city. So the main limitation that this literature review revealed is the choice of metric. Just relying on surface temperature is misleading because trees appear to have a better cooling effect then what most people actually

experience. Recent research in tropical regions gives evidence for humidity-mediated reversal which is otherwise concealed by surface temperature.

References

Borah, A., Datta, A., Kumar, A. S., Dave, R., & Bhatia, U. (2026). Dense canopies reverse the cooling effect of urban greening in humid cities. *Nature Communications, 17*, 5997. https://doi.org/10.1038/s41467-026-72636-w

Bowler, D. E., Buyung-Ali, L., Knight, T. M., & Pullin, A. S. (2010). Urban greening to cool towns and cities: A systematic review of the empirical evidence. *Landscape and Urban Planning, 97*(3), 147–155. https://doi.org/10.1016/j.landurbplan.2010.05.006

Chakraborty, T., Hsu, A., Manya, D., & Sheriff, G. (2019). Disproportionately higher exposure to urban heat in lower-income neighborhoods: A multi-city perspective. *Environmental Research Letters, 14*(10), 105003. https://doi.org/10.1088/1748-9326/ab3b99

Chen, B., Wu, S., Song, Y., Webster, C., Xu, B., & Gong, P. (2022). Contrasting inequality in human exposure to greenspace between cities of Global North and Global South. *Nature Communications, 13*, 4636. https://doi.org/10.1038/s41467-022-32258-4

Du, M., Li, N., Hu, T., Yang, Q., Chakraborty, T. C., Venter, Z., & Yao, R. (2024). Daytime cooling efficiencies of urban trees derived from land surface temperature are much higher than those for air temperature. *Environmental Research Letters, 19*(4), 044037. https://doi.org/10.1088/1748-9326/ad30a3

Guo, Z., Esperón-Rodríguez, M., Davin, E., Huang, H., Chen, B., Hejazi, M., Wu, J., Wang, J., Ge, Y., Song, G., Zhao, Y., Feng, K., Lin, C., Gong, P., & Zhou, Y. (2026). Global urban vegetation exhibits divergent thermal effects: From cooling to warming as aridity increases. *Science Advances, 12*(1), eaea9165. https://doi.org/10.1126/sciadv.aea9165

Ibsen, P. C., Borowy, D., Dell, T., Greydanus, H., Gupta, N., Hondula, D. M., Meixner, T., Santelmann, M. V., Shiflett, S. A., Sukop, M. C., Swan, C. M., Talal, M. L., Valencia, M., Wright, M. K., & Jenerette, G. D. (2021). Greater aridity increases the magnitude of urban

nighttime vegetation-derived air cooling. *Environmental Research Letters, 16*(3), 034011. https://doi.org/10.1088/1748-9326/abdf8a

Iungman, T., Cirach, M., Marando, F., Pereira Barboza, E., Khomenko, S., Masselot, P., Quijal-Zamorano, M., Mueller, N., Gasparrini, A., Urquiza, J., Heris, M., Thondoo, M., & Nieuwenhuijsen, M. (2023). Cooling cities through urban green infrastructure: A health impact assessment of European cities. *The Lancet, 401*(10376), 577–589. https://doi.org/10.1016/S0140-6736(22)02585-5

Kim, J., Khouakhi, A., Corstanje, R., & Johnston, A. S. A. (2024). Greater local cooling effects of trees across globally distributed urban green spaces. *Science of the Total Environment, 911*, 168494. https://doi.org/10.1016/j.scitotenv.2023.168494

Li, Y., Svenning, J.-C., Zhou, W., Zhu, K., Abrams, J. F., Lenton, T. M., Ripple, W. J., Yu, Z., Teng, S. N., Dunn, R. R., & Xu, C. (2024). Green spaces provide substantial but unequal urban cooling globally. *Nature Communications, 15*, 7108. https://doi.org/10.1038/s41467-024-51355-0

Manoli, G., Fatichi, S., Schläpfer, M., Yu, K., Crowther, T. W., Meili, N., Burlando, P., Katul, G. G., & Bou-Zeid, E. (2019). Magnitude of urban heat islands largely explained by climate and population. *Nature, 573*(7772), 55–60. https://doi.org/10.1038/s41586-019-1512-9

Meili, N., Manoli, G., Burlando, P., Carmeliet, J., Chow, W. T. L., Coutts, A. M., Roth, M., Velasco, E., Vivoni, E. R., & Fatichi, S. (2021). Tree effects on urban microclimate: Diurnal, seasonal, and climatic temperature differences explained by separating radiation, evapotranspiration, and roughness effects. *Urban Forestry & Urban Greening, 58*, 126970. https://doi.org/10.1016/j.ufug.2020.126970

Peng, S., Piao, S., Ciais, P., Friedlingstein, P., Ottle, C., Bréon, F.-M., Nan, H., Zhou, L., & Myneni, R. B. (2012). Surface urban heat island across 419 global big cities. *Environmental Science & Technology, 46*(2), 696–703. https://doi.org/10.1021/es2030438

Schwaab, J., Meier, R., Mussetti, G., Seneviratne, S., Bürgi, C., & Davin, E. L. (2021). The role of urban trees in reducing land surface temperatures in European cities. *Nature Communications, 12*, 6763. https://doi.org/10.1038/s41467-021-26768-w

Shah, A., Garg, A., & Mishra, V. (2021). Quantifying the local cooling effects of urban green spaces: Evidence from Bengaluru, India. *Landscape and Urban Planning, 209*, 104043. https://doi.org/10.1016/j.landurbplan.2021.104043

Venter, Z. S., Chakraborty, T., & Lee, X. (2021). Crowdsourced air temperatures contrast satellite measures of the urban heat island and its mechanisms. *Science Advances, 7*(22), eabb9569. https://doi.org/10.1126/sciadv.abb9569

Yang, Q., Huang, X., Tong, X., Xiao, C., Yang, J., Liu, Y., & Cao, Y. (2022). Global assessment of urban trees' cooling efficiency based on satellite observations. *Environmental Research Letters, 17*(3), 034029. https://doi.org/10.1088/1748-9326/ac4c1c

Yu, Z., Xu, S., Zhang, Y., Jørgensen, G., & Vejre, H. (2018). Strong contributions of local background climate to the cooling effect of urban green vegetation. *Scientific Reports, 8*, 6798. https://doi.org/10.1038/s41598-018-25296-w

Zhao, L., Lee, X., Smith, R. B., & Oleson, K. (2014). Strong contributions of local background climate to urban heat islands. *Nature, 511*(7508), 216–219. https://doi.org/10.1038/nature13462

Zhou, W., Wang, J., & Cadenasso, M. L. (2017). Effects of the spatial configuration of trees on urban heat mitigation: A comparative study. *Remote Sensing of Environment, 195*, 1–12. https://doi.org/10.1016/j.rse.2017.03.043